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How academic librarians combat student plagiarism

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ABSTRACT

Plagiarism is one of Academia's most widely discussed topics, as it is considered severe academic misconduct. Academic librarians can become pioneers on the path to academic integrity and play a tremendously important role in combating student plagiarism. Using a quantitative approach, the present study attempts to investigate academic libraries' plagiarism preventive and detection role from the perspective of public university librarians in Greece. For the study, a five-faceted conceptual framework for the role of librarians was developed and evaluated. Results indicate that librarians have actively taken the lead in raising awareness of plagiarism and providing plagiarism education, support, and counselling. Yet, their participation and contribution to developing institutional plagiarism strategies could have been higher than the other roles. Similarly, the results showed low participation of library professionals in plagiarism detection services, underscoring the need for effective collaboration with other academic stakeholders. Analysis of differences based on demographic factors suggests that individuals with higher levels of education are more closely involved in various roles and activities related to plagiarism management. Implications of the findings are discussed.

Introduction

Plagiarism is a complex and multidimensional phenomenon, the management of which involves many different factors and practices. To this end, academic institutions worldwide have established mechanisms to combat academic malpractice and to promote the ethical use of information (Duggan, 2009; Hendrix, 2016; Maurer et al., 2006). Within the Greek academic community, several higher education institutions (HEIs) are still striving to address the issue of student plagiarism -which by Law 4957/2022 is considered a disciplinary offence- focusing their endeavors on the development of anti-plagiarism strategies, policies, and procedures, in the form of regulations, guidelines, code of ethics, information literacy courses, plagiarism detection software and the use of institutional repositories (Kokkinaki et al., 2013; Kokkinos & Koulouris, 2020). Although academic libraries should be leading anti-plagiarism efforts (Datig & Russell, 2014; Han, 2017; Lampert, 2008; Michalak et al., 2018; Rodafinos, 2018), there is still limited empirical research on information scientists' views regarding the part academic libraries can play in combating academic misconduct (Bartlett & Casselden, 2011; Snyder Gibson & Chester-Fangman, 2011). Considering this, the current study seeks to explore Greek academic librarians':

(1) perceptions regarding their role and the role of libraries in combating plagiarism
(2) engagement with plagiarism prevention and detection activities
(3) collaboration efforts with the academic community to combat student plagiarism and
(4) challenges faced with respect to their active participation in tackling plagiarism in Academia.

Possible differences in the perceptions above based on demographic factors (age, work experience, and highest level of education attained) are also examined.

In the context of the current study, plagiarism is defined as "the act of appropriating the expressions or the wording and, consequently, the thoughts and ideas of someone else, without crediting the creator or without referencing the used source" (Kokkinos & Koulouris, 2020, p. 18).

Literature review

Legal, ethical, and educational implications of plagiarism in Academia

To protect the intellectual property of written work, the world's first

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copyright law, the British Statute of Anne, was enacted in 1710 (Feather, 1980), while at the same period, great emphasis was given to the moral offence of plagiarism (Goldgar, 2001). Since then, intellectual property laws and regulations have been implemented in most countries worldwide, reflecting the respective cultural and economic realities (Coombe & Turcotte, 2012). At the local level, Greek legislation has been harmonized with the European Union egal framework, which defends the moral rights of authors. Still, there is no separate legal framework regarding plagiarism (Tsolis, 2016). Regarding academic institutions, legal implications of plagiarism vary depending on the form and type of plagiarism, intention, academic integrity, ethical use of information and other rules governing research and scholarly practice (Saunders, 2010). This is mainly because plagiarism, in the form of copying text from someone else's work without giving them proper credits, is likely to have no financial impact on the author, while copyright focuses mainly on property rights and financial gain (Anderson, 2016; Haviland & Mullin, 2009).

The academic community also recognizes plagiarism as a growing ethical issue (Bennett et al., 2011; Saunders, 1993). In particular, plagiarism is considered one of the most common forms of academic dishonesty (Eret & Gokmenoglu, 2010; Jones, 2011) and misconduct (Awasthi, 2019; Park, 2003; Perry, 2010), a violation of academic ethics (Eret & Gokmenoglu, 2010) and a moral offence, posing an imminent threat to the academic integrity of HEIs (Boehm & Justice, 2009; Thomas & Sassi, 2011). As such, even mere allegations of plagiarism can damage the career of students, the credibility of faculty and the academic institution's reputation (Auer & Krupar, 2001; Snyder Gibson & Chester-Fangman, 2011; Thuku, 2017). Legal and moral implications aside, plagiarism is an educational issue (Gunnarsson et al., 2014) which may lead to student inability to achieve learning objectives (Darr, 2019) and has implications beyond the classroom (Thomas & Sassi, 2011). Ethical and moral values, personal and social responsibility, intellectual development, independent work, and integrity dilemmas are issues that students will face in professional settings (Association of American Colleges and Universities, 2007; Wong et al., 2016).

Causes of plagiarism

Although a general distinction can be made between intentional and unintentional plagiarism (Selemani et al., 2018), it is not always easy to pinpoint the causes of plagiarism (Burke, 2005). Relevant literature has identified a number of reasons for students' academic misconduct, which can be classified in twelve broad categories: (1) foreign language problems; (2) time constrains, poor time management or lack of time management skills; (3) academic pressure and high student workload; (4) ignorance or poor knowledge of what plagiarism is; (5) absence of consequences, penalties, and/or regulations and policies in academic institutions; (6) lack of adequate academic writing and research skills, including proper use of sources and proper ways of citing, referencing, quoting, paraphrasing, and summarizing and/or us of critical thinking; (7) difficulties in managing research process; (8) pressure or desire to achieve high levels of performance and good grades; (9) lack of respect to academic rules and to course requirements including losing interest in learning; (10) teaching factors including negative students' attitude towards faculty or a specific class; (11) factors related to student personality, values and attitudes or cultural issues; and (12) temptation and opportunities due to unlimited access to digital materials and/or to the wide use of essay writing web services (Anaman & Agyei, 2021; Davies & Howard, 2016; Devlin & Gray, 2007; Ismail et al., 2014; Jilani & Ahmad, 2021; Park, 2003; Roberts, 2007; Romanowski, 2022; Shirazi et al., 2010; Šprajc et al., 2017; Stephens & Nicholson, 2008; Thomas, 2004).

Plagiarism management strategies and the role of libraries

HEIs have adopted various mechanisms and practices to prevent and combat plagiarism, including: a) information literacy education; b) establishment of plagiarism advisory service centers; c) use of plagiarism detection tools and software (e.g., Turnitin), and d) application of plagiarism penalties (Duggan, 2009; Hendrix, 2016). However, anti-plagiarism policies have only sometimes been effective in managing academic misconduct (Prins & Lathrop, 2014), mainly because they were overly focused on suppression rather than prevention (Mundava & Chaudhuri, 2007). For this reason, Darr (2019) argues that anti-plagiarism strategies should not create a culture of fear among university students. Instead, academic institutions should develop clear policies and incorporate specific procedures, rules and mechanisms for plagiarism detection and prevention (Bretag, 2013; Glendinning, 2014).

In Greece, HEIs have incorporated anti-plagiarism policies in their internal regulations, following relevant laws and regulations. Thereby, all dissertations and research work of students should be submitted for plagiarism checking using anti-plagiarism tools. Academic libraries are usually responsible for managing plagiarism detection software. Additionally, many academic libraries provide information on plagiarism, academic dishonesty and research ethics and have created guides and other instructional material for students. At the same time, they have also been involved in delivering information literacy training (Kokkinos & Koulouris, 2020).

Overall, the literature suggests two distinct supportive roles of academic librarians in combating plagiarism, i.e., plagiarism educators and plagiarism detectors (Mansoor & Ameen, 2018). Recognizing the importance of collaborative efforts between librarians and all other HEIs stakeholders, the current study expands the proposition and depicts five supportive roles of librarian professionals, schematically presented in Fig. 1:

a. Librarians as plagiarism *informants*: raising student awareness about plagiarism and providing valuable information to prevent plagiarism.
b. Librarians as plagiarism *counsellors*: providing guidance, support, and counselling to students to avoid plagiarism.
c. Librarians as plagiarism *educators*: educating students about plagiarism and research ethics and contributing to students' development of information literacy skills.
d. Librarians as plagiarism *detectors*: supporting academic staff with student plagiarism detection services; and
e. Librarians as professional *policy advisors*: participating in developing institution plagiarism policy.

Research methodology

Study design

As previously mentioned, the study aimed to assess Greek library professionals' roles in combating plagiarism. To this end, a quantitative research strategy was deployed. More specifically, a cross-sectional research design was adopted, and primary data were collected via an online structured questionnaire. Survey has been the leading research method for primary data collection in Library & Information Science (LIS) for the last fifty years (Vander Kooy, 2020). It was adopted because it is a practical and cost-effective method for reaching large populations and usually yields large response rates (Gilbert, 2008; Regmi et al., 2017).

Sampling and data collection

The study population comprised 686 professionals employed in academic libraries attached to the 25 Greek public HEIs. The online questionnaire was sent to library administrators and staff via their emails and the Greek Academic Libraries Electronic Mailing List, known as Acadelib. The institutional ethics committee approved the survey, and all participants provided informed consent before participation.

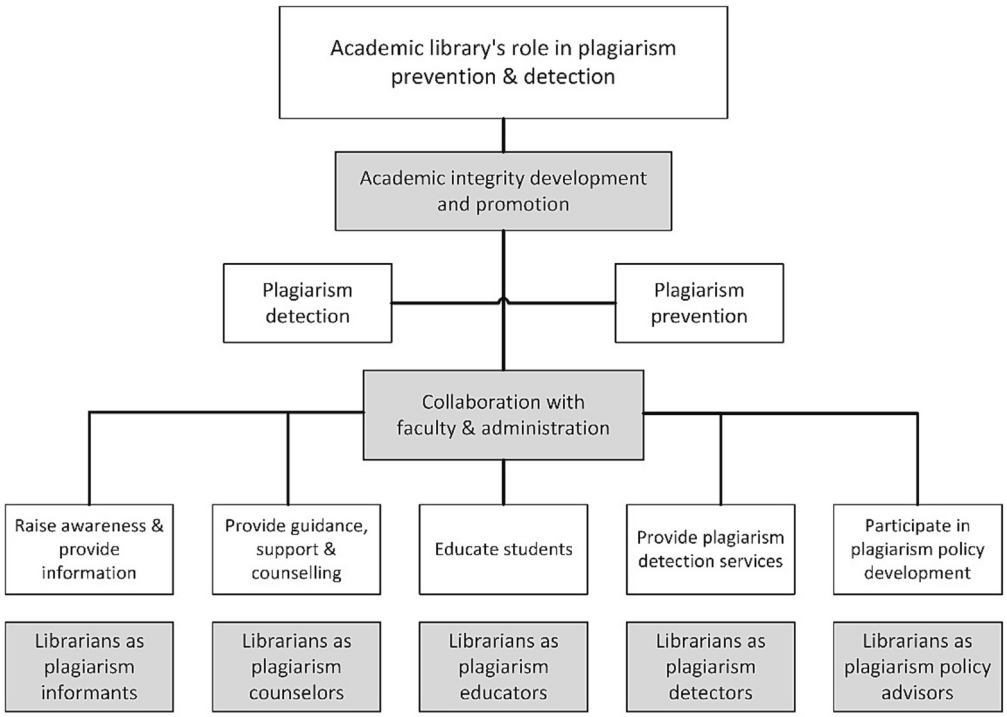


Fig. 1. A framework for academic library's role in the prevention and detection of plagiarism.

Overall, 147 questionnaires suitable for analysis were returned for an effective response rate of 21.5 %. The final sample (see Table 1) is nationally representative, as it includes responses from library employees working in 24 of the 25 Greek HEIs. Most participants were permanent employees (91 %) and worked in academic libraries for over 16 years (77 %). Regarding age, 69 % of respondents were over 46 years old, and no responses from individuals less than 36 years of age were obtained.

Table 1
Characteristics of the sample ($n = 147$).

Constructs	Indicators	Frequency	(%)
Employment type	Permanent	134	91
	Temporary	13	9
Library professionals	Librarians	135	92
	Other library professionals[a]	12	8
Age groups in years	18 to 25	0	0
	26 to 35	0	0
	36 to 45	46	31
	46 to 55	82	56
	Over 55	19	13
Academic library work experience in years	1 to 5	5	3
	6 to 10	9	6
	11 to 15	20	14
	16 to 20	30	20
	Over 20	83	56
Highest level of education completed	Secondary	2	1.4
	Bachelor's degree	72	49.0
	Master's degree	65	44.2
	PhD	8	5.4
Field of education	BA in Library and Information Science	117	76
	BA in Computer Science	2	1
	BA in other fields of study	12	8
	MSc in Library and Information Science	57	39
	MSc in Computer Science	9	6
	MSc in other fields of study	6	4
	PhD in Library and Information Science	6	4
	PhD in other fields of study	2	1

[a] Laboratory teaching staff, library assistants, technical support staff.

According to the latest available statistics from the Hellenic Statistical Authority (2021) regarding the number of individuals employed in Greek libraries, there is a mere 2.4 % increase between 2018 and 2020. The population distribution of the Greek academic library staff (a relatively aged population) and the low percentage of representation of younger age groups are directly related to the low number of hires in recent years, primarily due to the economic crisis. Other studies using Greek academic librarian samples provided similar age data (Alvanoudi, 2020; Koloniari & Fassoulis, 2017; Kouis et al., 2021; Kyprianos, 2022; Vraimaki et al., 2019). Finally, 92 % of the sample were librarians, 80 % hold a bachelor's degree in library and information science and 49 % had an MSc degree in LIS and other fields of study.

Questionnaire development and data analysis methodology

A structured online questionnaire was developed guided by an in-depth critical review of published scholarly research related to student plagiarism in Academia and libraries' role in combatting plagiarism. The main survey questions (see Appendix A) were organized into four sections that assessed librarians':

(1) perceptions of plagiarism
(2) perceptions regarding their role and the role of libraries in combating plagiarism
(3) current engagement with plagiarism prevention and detection activities
(4) challenges in the involvement of libraries in controlling plagiarism.

Answers to close-ended questions were given on a five-point Likert-type scale. At the same time, some of the questions also contained an additional free-text option so that participants could elaborate on their responses. Face validity was assessed by an expert panel of academic librarians with experience in research supervision and plagiarism policies; minor modifications were made based on their comments. Finally, data were analyzed using descriptive statistics, i.e., frequencies and median (MDN) and interquartile range (IQR) for central tendency, that

are appropriate for analyzing and interpreting ordinal data (Boone & Boone, 2012; Siegel, 1956). At a second stage, inferential tests were used to determine if there were statistically significant differences among respondents' opinions based on demographic characteristics, namely, age, education, and work experience. All analyses were conducted using IBM® SPSS® v28.

Results and discussion

Plagiarism perceptions: types of and reasons for plagiarism

Initially, library professionals' perceptions about plagiarism were assessed. The first step in preventing plagiarism is to identify the reasons, the forms, and the factors that cause the phenomenon's emergence. As such, distinguishing between unintentional and intentional plagiarism is critical when building a framework for preventing and detecting plagiarism incidents. Survey participants were asked to classify, according to their experience, plagiarism incidents as either unintentional or intentional. 63 % of the participants indicated that plagiarism is primarily inadvertent. The relevant literature also supports this finding (Eret & Gokmenoglu, 2010; Maurer et al., 2006; Selemani et al., 2018; Volkov et al., 2011).

Moreover, participants were asked to indicate the most frequently encountered types of plagiarism (see Table 2). Findings suggest that, although plagiarism can take many forms, direct plagiarism -i.e., copying text (words, expressions, sentences, paragraphs), media or ideas without (correctly) citing the source- is still the most common type of student plagiarism. This finding is consistent with results from the literature (Cooper, 2016; Park, 2003), where copy/pasting is the most dominant plagiarism type. However, "patchwriting" is considered an essential phase in the development of academic writing and citation skills of students early in their academic careers (Buranen, 2009; Burger, 2018).

Library employees were also asked about the reasons students plagiarize. Results, presented in Table 3, indicate that the two principal reasons for plagiarism are lack of adequate academic writing, research, and critical thinking skills and ignorance or poor knowledge of what constitutes plagiarism. Two other common reasons are temptation, easy access to online materials, and difficulties managing the research process. Indeed, digital plagiarism is one of the most popular types today (Darr, 2019). 67 % of the participants also noted the absence of plagiarism regulations and policies and the nonexistence of penalties, suggesting the need to strengthen plagiarism detection and develop more explicit academic integrity policies.

Table 2
Types of student plagiarism.

	N	R	S	O	VO	MDN	IQR
Copying text (words, expressions, sentences, paragraphs) or ideas without (correctly) citing the author or the source	2	4	27	72	42	4	1
Copying and pasting segments of text without quotation marks, even if a citation referring has been added	2	4	37	71	33	4	1
Paraphrasing or summarizing text without (correctly) citing the author or the source	2	2	35	73	35	4	1
Submitting another student's work (e.g., assignment or paper) with or without her/his knowledge of permission	8	55	55	24	5	3	1
Submitting or presenting a group work assignment as individual academic work	11	52	55	23	6	3	1
Buying an essay or other written academic work from professional essay writing services or other third parties	6	33	49	38	21	3	2
Copying from other students or collaborating with other students without the lecturer's approval during exams or classroom assessments	8	24	60	38	17	3	1
Copying a substantial amount of content from a source, regardless of whether there is proper citation and referencing of the source	3	20	67	46	11	3	1
Using incorrect citation or bibliographic referencing, which does not provide accurate information about the original sources	3	15	51	50	28	4	1
Self-plagiarism in terms of presenting or reusing any of their previous (published) work or assignment without attribution	8	39	49	41	10	3	2
Copying media (images, audio, or video) from other sources and presenting them in their own academic work or project without attribution	6	21	42	45	33	4	1

Notes: N – Never, R - Rarely, S – Sometimes, O - Often, VO – Very Often, MDN – Median, IQR – Interquartile range.

The role of librarians in plagiarism prevention and detection

Participants were asked to assess their agreement level with the academic library's five plagiarism-combating roles (see Fig. 1). The results (Table 4) indicate that participants view themselves primarily as informants, counsellors, and educators and less as "detectors" and policy advisors. They utterly agree that the library should play an active and essential role in students' awareness, support, and education to prevent plagiarism incidents. In contrast, engagement with plagiarism detection measures has been reported as a minor concern. These views are also evident in the relevant literature, where several scholars and library professionals argue that academic library staff not only can contribute to the prevention of plagiarism but also has the potential to support academics in detecting and identifying student plagiarism incidents (Burke, 2005; Idiegbeyan-ose et al., 2018; Jilani & Ahmad, 2021; Mansoor & Ameen, 2018; Snyder Gibson & Chester-Fangman, 2011) while others question librarians' "plagiarism detector" role (Bartlett & Casselden, 2011; Giersberg, 2012).

One-way ANOVA tests were used to examine the effects of demographic factors (age, highest level of education attained, and work experience) on participants' perceptions regarding the role of academic libraries in combating plagiarism. If the results were statistically significant, the Fisher's Least Significance Difference (LSD) (Fisher, 1936) post-hoc test was performed to determine which of the differences between the groups were significant.

According to the results, there are no statistically significant differences in participants' perceptions about the role of academic libraries in combating plagiarism based on their age, as opposed to work experience and educational level. Specifically, the results ($F = 2.575$, $df = 4$, sig. $= 0.041$) showed that participants with less than six years of professional experience agreed to a lesser extent with the role of the academic library in preventing student plagiarism through awareness and information ($M = 3.60$, $SD = 1.140$), in contrast to the participants with more library experience who fully agreed with this role (see Tables 5 and 6).

Results also indicated that the higher the educational background, the greater the level of agreement to engage in the proposed roles. Specifically, statistically significant results were found for the following two roles: librarians as plagiarism informants ($F = 3.680$, $df = 2$, sig. $= 0.028$) and librarians as plagiarism educators ($F = 2.428$, $df = 2$, sig. $= 0.018$), as seen in Tables 7 and 8.

Table 3
Reasons for student plagiarism.

	N	R	S	O	VO	MDN	IQR
Foreign language problems	4	19	57	44	23	3	1
Time constraints, poor time management or lack of time management skills	2	14	46	71	14	4	1
Academic pressure and high student workload	4	16	50	64	13	4	1
Unawareness or poor knowledge of the concept and the forms of plagiarism	1	12	23	62	49	4	1
Lack of plagiarism implications or penalties and/or absence of plagiarism regulations and policy of the academic institution	3	7	38	65	34	4	1
Lack of adequate academic writing and research skills, including the proper use of sources and the proper ways of citing, referencing, quoting, paraphrasing, and summarizing and/or use of critical thinking	1	12	19	63	52	4	1
Difficulties in managing the research process	1	6	36	66	38	4	2
Pressure or desire to achieve high levels of performance and good grades	1	13	42	67	24	4	1
Lack of respect for academic rules and course requirements, including losing interest in learning	3	29	52	53	10	3	1
Teaching factors, including negative students' attitudes towards academic and teaching staff or class	6	46	61	30	4	3	1
Factors related to student personality, personal values and attitudes or cultural issues	5	22	62	41	17	3	1
Temptation and opportunities due to unlimited access to digital materials and/or to the wide use of essay writing web services	2	4	31	71	39	4	2

Notes: N – Never, R - Rarely, S – Sometimes, O - Often, VO – Very Often, MDN – Median, IQR – Interquartile range.

Table 4
Aspects of librarians' role in combating plagiarism.

	SD	D	N	A	SA	MDN	IQR
Prevention of student plagiarism through awareness and provision of information	0	1	8	60	66	4	1
Provision of student support, counselling, and guidance to avoid plagiarism	0	3	7	57	68	5	1
Promotion of information literacy skills and ethical use of information through plagiarism education instruction	0	4	7	52	72	5	1
Plagiarism detection in students' written academic work	13	22	28	35	37	4	3
Participation in the development of plagiarism and academic integrity institutional policy	0	5	24	53	53	4	1

Notes: SD – Strongly disagree, D - Agree, N – Neutral, A - Agree, SA – Strongly Agree, MDN – Median, IQR – Interquartile range.

Current practice for the prevention and detection of plagiarism

Prevention activities

Survey participants were asked to indicate which plagiarism prevention activities they are currently engaged in and to what extent. The findings (see Table 9) show a significant variance in responses, with the recommendation of resources being the most popular activity, followed by providing information on copyright issues, academic integrity, the ethical use of information and the provision of citation and referencing support. The overall results show that library professionals, as custodians of information, are in a unique position to raise student awareness regarding plagiarism and ethical use of information, findings confirmed by recent literature (Darr, 2019; Jilani & Ahmad, 2021; Mansoor & Ameen, 2018).

Table 5
Aspects of librarians' role in combating plagiarism by work experience.

One – Way ANOVA						
	Work experience (years)	n	Mean	SD	F	Sig.
Prevention of student plagiarism through awareness and provision of information	1–5	5	3.60	1.140	2.575	0.041[a]
	6–10	7	4.71	0.488		
	11–15	15	4.47	0.516		
	16–20	29	4.45	0.632		
	Over 20	79	4.42	0.612		
Provision of student support, counselling, and guidance to avoid plagiarism	1–5	5	4.20	0.837	1.167	0.328
	6–10	7	4.57	0.535		
	11–15	15	4.73	0.458		
	16–20	29	4.34	0.670		
	Over 20	79	4.37	0.737		
Promotion of information literacy skills and ethical use of information through plagiarism education instruction	1–5	5	4.00	1.225	1.292	0.277
	6–10	7	4.43	0.787		
	11–15	15	4.73	0.458		
	16–20	29	4.31	0.604		
	Over 20	79	4.43	0.763		
Plagiarism detection in students' written academic work	1–5	5	3.80	1.095	0.579	0.678
	6–10	7	3.86	1.345		
	11–15	15	3.73	1.280		
	16–20	29	3.28	1.222		
	Over 20	79	3.41	1.364		
Participation in the development of plagiarism and academic integrity institutional policy	1–5	5	3.60	1.140	0.663	0.619
	6–10	7	4.14	1.215		
	11–15	15	4.07	0.961		
	16–20	29	4.10	0.724		
	Over 20	79	4.20	0.807		

[a] Significant at 0.05 level.

Participants were also asked to indicate the plagiarism awareness raising methods and activities they are engaged in. Results presented in Table 10 suggest that library professionals mainly provide information about plagiarism at the library's help desk. Notably, all other options received relatively low scores, suggesting that there is limited variety in the actions taken to raise plagiarism awareness by academic libraries. Indeed, recent studies in this area suggest that academic libraries should promote their role as anti-plagiarism information centers through awareness campaigns using a variety of methods and platforms (websites, posters, social media) (Adjoa Yebowaah & Sanche, 2021; Idiegbeyan-ose et al., 2018).

Most responses related to activities on plagiarism awareness raising methods, which fell into the 'other' category, concerned conducting seminars offered to all library members or upon request (on demand) for specific groups. In some cases, library staff co-organized seminars with faculty staff and/or offered interactive workshops on using the library's resources and developed corresponding material as part of electronic courses.

Regarding demographic variables, age and work experience did not appear to influence librarians' engagement in plagiarism prevention. On the contrary, differences were depicted in plagiarism prevention based on participants' educational level. As shown in Tables 11–12, the higher the educational level, the higher the engagement in plagiarism prevention activities. Specifically, there is a statistically significant difference between participants with a bachelor's degree and those with a postgraduate degree in their current plagiarism prevention practices,

Table 6
Post-hoc test for ANOVA (Table 5) statically significant results.

Multiple comparisons (LSD Post-Hoc test)							
Dependent variable	(I) Work experience (years)	(J) Work experience (years)	Mean difference (I-J)	Std. error	Sig.	95 % confidence interval	
						Lower bound	Upper bound
Prevention of student plagiarism through awareness and provision of information	1–5	6–10	−1.114[a]	0.366	0.003	−1.84	−0.39
		11–15	−0.867[a]	0.323	0.008	−1.50	−0.23
		16–20	−0.848[a]	0.303	0.006	−1.45	−0.25
		Over 20	−0.818[a]	0.288	0.005	−1.39	−0.25

[a] Mean difference is significant at the 0.05 level.

Table 7
Aspects of librarians' role in combating plagiarism by. level of education.

One – Way ANOVA test						
	Highest level of education attained	n	Mean	SD	F	Sig.
Prevention of student plagiarism through awareness and provision of information	Bachelor	64	4.30	0.659	3.680	0.028[a]
	Master	62	4.53	0.535		
	PhD	8	4.75	0.463		
Provision of student support, counselling, and guidance to avoid plagiarism	Bachelor	64	4.30	0.770	2.428	0.092
	Master	62	4.50	0.594		
	PhD	8	4.75	0.463		
Promotion of information literacy skills and ethical use of information through plagiarism education instruction	Bachelor	64	4.28	0.786	4.128	0.018[a]
	Master	62	4.55	0.592		
	PhD	8	4.88	0.354		
Plagiarism detection in students' written academic work	Bachelor	64	3.38	1.241	0.385	0.681
	Master	62	3.52	1.388		
	PhD	8	3.75	1.282		
Participation in the development of plagiarism and academic integrity institutional policy	Bachelor	64	4.08	0.822	1.055	0.351
	Master	62	4.19	0.846		
	PhD	8	4.50	0.535		

[a] Significant at 0.05 level.

with the former less engaged in activities related to the provision of advice and anti-plagiarism techniques (F = 4.389, df = 2, sig. = 0.014).

Education activities

Next, library professionals were asked to indicate the main topics covered in the anti-plagiarism instruction and the basic instruction delivery methods. Regarding topics (see Table 13), results showed that most commonly training involves digital information sources

Table 9
Engagement with plagiarism prevention activities.

	N	R	S	O	VO	MDN	IQR
Recommend information sources to students on their topic of interest	7	8	28	41	39	4	2
Answer questions about plagiarism-related issues	21	25	35	24	18	3	2
Provide information on copyright issues, academic integrity, and the ethical use of information	9	15	34	38	27	4	1
Inform about the consequences of plagiarism	14	29	29	31	20	3	2
Inform about institutional policy and procedures	21	26	33	29	14	3	2
Provide support to improve citation and referencing skills	13	18	30	29	33	4	3
Give advice on how to avoid plagiarism with the provision of anti-plagiarism techniques	20	27	26	25	25	3	2
Assist in the preparation of assignments/essays	27	23	24	29	20	3	2
Provide support to students of a suspected plagiarism incident	46	32	22	11	12	2	2

Notes: N – Never, R - Rarely, S – Sometimes, O - Often, VO – Very Often, MDN – Median, IQR – Interquartile range.

Table 8
Post-hoc test for ANOVA (Table 7) statically significant results.

Multiple comparisons (LSD Post-Hoc test)							
Dependent variable	(I) Highest level of education attained	(J) Highest level of education attained	Mean difference (I-J)	Std. error	Sig.	95 % confidence interval	
						Lower bound	Upper bound
Prevention of student plagiarism through awareness and provision of information	Bachelor	Master	−0.235[a]	0.106	0.028	−0.44	−0.03
		PhD	−0.453[a]	0.223	0.044	−0.89	−0.01
	Master	Bachelor	0.235[a]	0.106	0.028	0.03	0.44
		PhD	−0.218	0.223	0.331	−0.66	0.22
	PhD	Bachelor	0.453[a]	0.223	0.044	0.01	0.89
		Master	0.218	0.223	0.331	−0.22	0.66
Promotion of information literacy skills and ethical use of information through plagiarism education instruction	Bachelor	Master	−0.267[a]	0.122	0.030	−0.51	−0.03
		PhD	−0.594[a]	0.256	0.022	−1.10	−0.09
	Master	Bachelor	0.267[a]	0.122	0.030	0.03	0.51
		PhD	−0.327	0.257	0.206	−0.83	0.18
	PhD	Bachelor	0.594[a]	0.256	0.022	0.09	1.10
		Master	0.327	0.257	0.206	−0.18	0.83

[a] Mean difference is significant at the 0.05 level.

Table 10
Awareness-raising methods.

	Frequency	Percentage (%)
Library help desk	82	66.67
Educational material on the library's website/portal	47	38.21
Phone calls	46	37.40
Information and resources on the library's website/ portal	41	33.33
Online communication through digital library services (e.g., live chats)	32	26.02
Emails	28	22.76
Use of social media (marketing) tools	24	19.51
Discussion forums and newsgroups	8	6.50
Other traditional marketing techniques (flyers, posters, brochures etc.)	5	4.07
None of the above	20	16.26
Other	11	8.94

Note: Multiple answers were allowed.

Table 11
Current practice for the prevention of plagiarism by educational level.

One – Way ANOVA test						
	Highest level of education attained	n	Mean	SD	F	Sig.
Recommend information sources to students on their topic of interest	Bachelor	58	3.84	1.056	0.260	0.772
	Master	58	3.81	1.146		
	PhD	122	3.81	1.108		
Answer questions about plagiarism-related issues	Bachelor	58	2.69	1.217	2.564	0.081
	Master	58	3.22	1.312		
	PhD	6	3.00	1.414		
Provide information on copyright issues, academic integrity, and the ethical use of information	Bachelor	58	3.31	1.079	1.768	0.175
	Master	58	3.64	1.238		
	PhD	6	4.00	0.894		
Inform about the consequences of plagiarism	Bachelor	58	2.86	1.131	2.717	0.070
	Master	58	3.40	1.350		
	PhD	6	3.17	0.983		
Inform about institutional policy and procedures	Bachelor	58	2.62	1.105	3.415	0.036
	Master	58	3.21	1.281		
	PhD	6	3.17	1.835		
Provide support to improve citation and referencing skills	Bachelor	58	3.26	1.292	1.031	0.360
	Master	58	3.60	1.324		
	PhD	6	3.50	1.049		
Give advice on how to avoid plagiarism with the provision of anti-plagiarism techniques	Bachelor	58	2.71	1.311	4.389	0.014[a]
	Master	58	3.43	1.378		
	PhD	6	3.33	1.033		
Assist in the preparation of assignments/ essays	Bachelor	58	2.79	1.373	0.717	0.490
	Master	58	3.10	1.423		
	PhD	6	3.00	1.414		
Provide support to students of a suspected plagiarism incident	Bachelor	58	2.02	1.192	2.383	0.097
	Master	58	2.53	1.417		
	PhD	6	2.50	1.049		

[a] Significant at 0.05 level.

management and/or citation instruction, followed by project and research data management tools training and/or information literacy instruction. Regarding delivery methods, seminars and/or workshops, and guides and manuals are the two most popular responses (see Table 14). The role of librarians and information scientists has been broadly associated with providing education and training in the ethical use of information and supporting the development of students' information literacy skills (Han, 2017; Lampert, 2004, 2008; Nilsen, 2012; Snyder Gibson & Chester-Fangman, 2011; Spring & Adams, 2013) and the findings of this research study support this. Finally, it is widely accepted that the best strategy for plagiarism prevention is student education (Darr, 2019; Stabingis et al., 2014).

A very small percentage of respondents (n = 4, 3 %) reported engaging in educational activities through other methods, such as Skype, providing information to students in person when asked, or informing teaching staff about Turnitin use.

Regarding differences based on demographics, educational background is the only factor influencing librarians' participation in plagiarism education activities. Like previous results, the higher the academic level, the higher the engagement in plagiarism educational activities. Participants with a bachelor's degree showed lower engagement in plagiarism education activities than librarians with graduate degrees (Tables 15–16). Activities such as information literacy instruction ($F = 4.708$, $df = 2$, sig. $= 0.011$), citation instruction ($F = 3.850$, $df = 2$, sig. $= 0.024$), research methods instruction ($F = 4.908$, $df = 2$, sig. $= 0.009$), and training on research data management tools ($F = 6.306$, $df = 2$, sig. $= 0.002$) show significant statistical differences between groups.

Detection activities

Participants were also asked to indicate their level of involvement in plagiarism detection. Results, presented in Table 17, suggest that librarians have a limited role in plagiarism detection. Respondents commented that plagiarism checking is a faculty responsibility, while the library is charged only with the administration of the repository. However, to enhance the efforts of higher education institutions to combat plagiarism, many scholars suggest that librarians can assist in the detection of plagiarism by students (Buranen, 2009; Idiegbeyan-ose et al., 2018; Jilani & Ahmad, 2021; Mansoor & Ameen, 2018; Snyder Gibson & Chester-Fangman, 2011). Especially with the widespread use of institutional repositories in academic institutions and their management by academic libraries, the role of librarians and information scientists should not be limited to the educational process (Burke, 2005) but can be extended to plagiarism detection for material submitted to institutional repositories. Most institutional repositories, as a form of research and scholarly publishing model for academic institutions (Moruf et al., 2019), are managed by academic libraries. At the national level, the results of a recent study (Kyprianos & Lygnou, 2022) showed that most academic libraries in Greece offer the provision and management of the institutional repository as part of their services.

Results of the ANOVA analysis did not indicate any differences based on the demographic variables examined.

Collaborative efforts

Regarding collaboration between the library and other university stakeholders, the study results (see Table 18) are somewhat disappointing, as most respondents have never partaken in synergistic activities. Collaboration among library professionals, academic teaching staff, academic departments, faculty, and administrators of higher education institutions is considered an essential factor in combating student plagiarism in Academia (Darr, 2019; Lampert, 2008; Madray, 2007; Morrow, 2022; Sciammarella, 2009). Therefore, communication efforts between library professionals and the academic community should be encouraged.

In addition, the results of ANOVA analysis did not show any differences based on the demographic variables examined for the collaboration between library professionals and the academic community in addressing student plagiarism in Academia.

Table 12
Post-hoc test for ANOVA (Table 11) statically significant results.

Multiple comparisons (LSD Post-Hoc test)							
Dependent variable	(I) Highest level of education attained	(J) Highest level of education attained	Mean difference (I-J)	Std. error	Sig.	95 % confidence interval	
						Lower bound	Upper bound
Give advice on how to avoid plagiarism with the provision of anti-plagiarism techniques	Bachelor	Master	−0.724[a]	0.248	0.004	−1.21	−0.23
		PhD	−0.626	0.572	0.276	−1.76	0.51
	Master	Bachelor	0.724[a]	0.248	0.004	0.23	1.21
		PhD	0.098	0.572	0.865	−1.03	1.23
	PhD	Bachelor	0.626	0.572	0.276	0–0.51	1.76
		Master	−0.098	0.572	0.865	−1.23	1.03

[a] Mean difference is significant at the 0.05 level.

Table 13
Anti-plagiarism instruction topics.

	N	R	S	O	VO	MDN	IQR
Instruction on academic ethical code and the ethical use of information	32	24	27	20	20	3	3
Information literacy instruction	30	20	28	21	24	3	2
Citation instruction (teaching of citation methods, bibliography, and in-text referencing)	23	19	25	33	23	3	2
Academic writing training, including paraphrasing and summarizing	46	26	24	17	10	2	2
Teaching research methods	24	29	27	26	17	3	2
Teaching (digital) source information management, including research database searching skills, information retrieval and synthesis	18	16	25	35	29	4	2
Project and research data management tools training	22	26	27	26	22	3	2

Notes: N – Never, R - Rarely, S – Sometimes, O - Often, VO – Very Often, MDN – Median, IQR – Interquartile range.

Table 14
Anti-plagiarism instruction delivery methods.

	Frequency	Percentage (%)
Workshops or seminars (synchronous method of education)	63	51.22
Guides and manuals	46	37.40
Lectures	29	23.58
Instructional/educational material and workbooks	22	17.89
Webinars, web tutorials or videos (asynchronous method of education)	14	11.38
Semester course	13	10.57
Course Management Systems (CMSs)	6	4.,88
Online discussion boards or blogs	0	0.00
None of the above	38	30.89
Others	4	3.25

Note: Multiple answers were allowed.

Challenges in combating plagiarism

Lastly, respondents were asked about their major challenges in combating plagiarism (see Table 19). Not surprisingly, more than half of the participants indicated collaboration with faculty and university administration as a significant hurdle. Respondents also mentioned that their institution had made no provision for the library's involvement. Interestingly, nearly half of the respondents believe they lack adequate knowledge and/or skills to educate students. Time constraints were also identified as a significant hindering factor. These findings are consistent with other relevant studies that emphasize the need for support from academic institution policies to enable libraries to more effectively do their multifaceted and vital work of deterring, preventing, and detecting plagiarism (Jilani & Ahmad, 2021; Mansoor & Ameen, 2018; Michalak et al., 2018; Snyder Gibson & Chester-Fangman, 2011).

Finally, results regarding differences depending on age and work experience were not statistically significant, contrary to educational background (Table 20). Particularly, participants with bachelor's degrees indicated a lack of adequate knowledge and/or skills to combat plagiarism to a greater extent than participants with higher educational levels (Table 21).

Table 15
Plagiarism education activities by educational level.

One – Way ANOVA Test						
	Highest level of education attained	n	Mean	SD	F	Sig.
Instruction on academic ethical code and the ethical use of information	Bachelor	58	2.52	1.466	2.109	0.126
	Master	58	3.05	1.290		
	PhD	6	2.83	1.835		
Information literacy instruction	Bachelor	58	2.52	1.405	4.708	0.011[a]
	Master	58	3.29	1.389		
	PhD	6	3.33	1.506		
Citation instruction (teaching of citation methods, bibliography, and in-text referencing)	Bachelor	58	2.78	1.415	3.850	0.024[a]
	Master	58	3.45	1.273		
	PhD	6	3.50	1.378		
Academic writing training, including paraphrasing and summarizing	Bachelor	58	2.10	1.252	2.686	0.072
	Master	58	2.52	1.301		
	PhD	6	3.17	1.835		
Teaching research methods	Bachelor	58	2.50	1.301	4.908	0.009[a]
	Master	58	3.19	1.235		
	PhD	6	3.50	1.643		
Teaching (digital) source information management, including research database searching skills, information retrieval and synthesis	Bachelor	58	3.07	1.387	2.655	0.074
	Master	58	3.64	1.238		
	PhD	6	3.33	1.633		
Project and research data management tools training	Bachelor	58	2.62	1.335	6.306	0.002[a]
	Master	58	3.29	1.311		
	PhD	6	4.17	0.753		

[a] Significant at 0.05 level.

Table 16
Post-hoc test for ANOVA (Table 15) statically significant results.

Multiple comparisons (LSD Post-Hoc test)							
Dependent variable	(I) Highest level of education attained	(J) Highest level of education attained	Mean difference (I-J)	Std. error	Sig.	95 % confidence interval	
						Lower bound	Upper bound
Information literacy instruction	Bachelor	Master	−0.776[a]	0.260	0.003	−1.29	−0.26
		PhD	−0.816	0.601	0.177	−2.01	0.37
	Master	Bachelor	0.776[a]	0.260	0.003	0.26	1.29
		PhD	−0.040	0.601	0.947	−1.23	1.15
	PhD	Bachelor	0.816	0.601	0.177	−0.37	2.01
		Master	0.040	0.601	0.947	−1.15	1.23
Citation instruction	Bachelor	Master	−0.672[a]	0.250	0.008	−1.17	−0.18
		PhD	−0.724	0.578	0.212	−1.87	0.42
	Master	Bachelor	0.672[a]	0.250	0.008	0.18	1.17
		PhD	−0.052	0.578	0.929	−1.20	1.09
	PhD	Bachelor	0.724	0.578	0.212	−0.42	1.87
		Master	0.052	0.578	0.929	−1.09	1.20
Teaching research methods	Bachelor	Master	−0.690[a]	0.239	0.005	−1.16	−0.22
		PhD	−1.000	0.552	0.072	−2.09	0.09
	Master	Bachelor	0.690[a]	0.239	0.005	0.22	1.16
		PhD	−0.310	0.552	0.575	−1.40	0.78
	PhD	Bachelor	1.000	0.552	0.072	−0.09	2.09
		Master	0.310	0.552	0.575	−0.78	1.40
Project and research data management tools training	Bachelor	Master	−0.672[a]	0.242	0.006	−1.15	−0.19
		PhD	−1.546[a]	0.559	0.007	−2.65	−0.44
	Master	Bachelor	0.672[a]	0.242	0.006	0.19	1.15
		PhD	−0.874	0.559	0.121	−1.98	0.23
	PhD	Bachelor	1.546[a]	0.559	0.007	0.44	2.65
		Master	0.874	0.559	0.121	−0.23	1.98

[a] Mean difference is significant at the 0.05 level.

Table 17
Plagiarism detection activities.

	N	R	S	O	VO	MDN	IQR
Checking assignments	99	9	5	6	3	1	0
Checking materials submitted to the institutional repository	95	9	8	2	8	1	0

Notes: N – Never, R - Rarely, S – Sometimes, O - Often, VO – Very Often, MDN – Median, IQR – Interquartile range.

Table 18
Collaborative efforts.

	N	R	S	O	VO	MDN	IQR
Meetings with faculty	73	27	10	9	2	1	1
Collaboration with faculty for anti-plagiarism courses	63	24	17	9	8	1	2
Collaboration with faculty for the development of anti-plagiarism assignments	66	28	14	8	5	1	1
Provide support for the detection of plagiarism	60	24	19	12	6	2	2
Provide instruction to faculty on plagiarism and the ethical use of information	61	24	19	11	6	1	2
Participation in the development of the plagiarism policy	77	20	14	8	2	1	1

Notes: N – Never, R - Rarely, S – Sometimes, O - Often, VO – Very Often, MDN – Median, IQR – Interquartile range.

Conclusions

Summary and implications

Plagiarism is inextricably bound to the learning process from the early stages of education. Being able to avoid plagiarism is a skill that builds gradually through education, practice, and experience, making it a phenomenon that is impossible to eliminate. Student plagiarism,

Table 19
Challenges in combating plagiarism.

	Frequency	Percentage (%)
Collaboration issues with faculty and administration	62	51.24
Lack of adequate knowledge and/or skills	52	42.98
Time constraints	49	40.50
Institutional policy for library involvement in plagiarism controlling	45	37.19
Communication difficulties with students	28	23.14
Other	5	4.13

Note: Multiple answers were allowed.

Table 20
Challenges in combating plagiarism by educational level.

One – Way ANOVA test						
	Highest level of education attained	n	Mean	SD	F	Sig.
Collaboration issues with faculty and administration	Bachelor	58	1.47	0.503	0.717	0.490
	Master	56	1.55	0.502		
	PhD	6	1.67	0.516		
Lack of adequate knowledge and/or skills	Bachelor	58	1.55	0.502	4.127	0.019[a]
	Master	56	1.32	0.471		
	PhD	6	1.17	0.408		
Time constraints	Bachelor	58	1.43	0.500	0.156	0.856
	Master	56	1.39	0.493		
	PhD	6	1.33	0.516		
Institutional policy for library involvement in plagiarism controlling	Bachelor	58	1.34	0.479	0.344	0.710
	Master	56	1.39	0.493		
	PhD	6	1.50	0.548		
Communication difficulties with students	Bachelor	58	1.17	0.381	1.997	0.140
	Master	56	1.27	0.447		
	PhD	6	1.50	0.548		

[a] Significant at 0.05 level.

Table 21
Post-hoc test for ANOVA (Table 20) statically significant results.

Multiple comparisons (LSD Post-Hoc test)							
Dependent variable	(I) Highest level of education attained	(J) Highest level of education attained	Mean difference (I-J)	Std. error	Sig.	95 % confidence interval	
						Lower bound	Upper bound
Lack of adequate knowledge and/ or skills	Bachelor	Master	0.230[a]	0.091	0.012	0.05	0.41
		PhD	0.385	0.208	0.066	−0.03	0.80
	Master	Bachelor	−0.230[a]	0.091	0.012	−0.41	−0.05
		PhD	0.155	0.208	0.458	−0.26	0.57
	PhD	Bachelor	−0.385	0.208	0.066	−0.80	0.03
		Master	−0.155	0.208	0.458	−0.57	0.26

[a] Mean difference is significant at the 0.05 level.

however, threatens both the effectiveness of the pedagogical endeavor and the professional integrity of HEIs (Burger, 2018; Coughlin, 2015). Therefore, academic institutions worldwide have established various strategies to prevent students from plagiarizing and promote the ethical use of information, and academic libraries can play a pivotal role in HEIs' efforts to maintain academic integrity.

In this context, the current study aimed to assess Greek library professionals' views on the role academic libraries can play in combating student plagiarism. Overall, the results showed that academic libraries should be placed at the forefront of student plagiarism prevention, with library professionals assuming the roles of informants, educators, and counsellors within these initiatives. In particular, it was found that academic libraries dynamically support plagiarism awareness and education efforts and lead information literacy instruction. However, librarians are not usually involved in policy development, and Greek HEIs have yet to include the library in their anti-plagiarism planning actively. Similarly, libraries have a minor role in plagiarism detection, indicating a significant cooperation deficit between libraries and the other HEI internal stakeholders.

Analysis of differences in respondents' opinions based on demographic factors suggests that individuals with higher levels of education are more closely involved in various roles and activities related to plagiarism management. In particular, the lower the educational level of participants, the less willing they are to engage in plagiarism prevention and education activities. Librarians with postgraduate degrees seem more committed to their plagiarism-combating roles and show higher engagement in plagiarism prevention and education practices. Finally, respondents, especially those in lower educational levels, feel they lack the necessary knowledge and skills to fulfil their plagiarism detection and prevention roles. These findings may be partially explained by the fact that individuals with a more active, personal involvement in research (e.g., for their postgraduate and doctoral theses) may be more sensitive to and experienced in matters of academic misconduct.

Library leaders that seek to strengthen the role of their libraries in plagiarism detection and prevention should target their efforts towards effectively communicating the knowledge, skills and services librarians can contribute to this end. Emphasis needs to be placed on outreach and collaboration initiatives that can establish in the minds of the academic community the central role libraries can play in maintaining academic integrity. In any case, planning should be based on solid evidence about the library's current level of involvement in the parent institution's anti-plagiarism strategy. The extended five-facet framework of librarians' role in supporting HEIs' anti-plagiarism initiatives proposed by the recent research can provide library managers with a solid base for conducting this assessment. Lastly, library leaders should pay attention to training provisions to keep up with the technological developments, such as AI chatbots, that have created a new battleground for academic institutions (Khalil & Er, 2023).

Limitations and further research

A potential limitation of the study is that survey participants were not given specific instructions on the differences between Likert response options. Thereafter, the responses were open to personal interpretation, which might have had a negative impact on the results. Also, the study would benefit from a larger sample of library professionals in the current academic context. Moreover, it would be valuable to enhance findings with qualitative data (from in-depth interviews or focus groups) to understand better the practices and views described. Additionally, as the survey population was limited to librarians employed in Greek academic libraries, similar studies can be conducted in other educational institutions worldwide to shed light on the crucial roles of librarians in combating student plagiarism. Finally, further research should also explore collaboration initiatives between library professionals and other academic stakeholders as it is a critical factor in hindering the full deployment of librarians' knowledge and skills in plagiarism prevention.

CRediT authorship contribution statement

Venetia Giannakouli: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration. **Eftichia Vraimaki:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration. **Alexandros Koulouris:** Conceptualization, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Dionysis Kokkinos:** Methodology, Investigation, Resources, Writing – original draft, Writing – review & editing, Visualization. **Dimitrios Kouis:** Software, Validation, Writing – review & editing, Visualization. **Konstantinos Kyprianos:** Resources, Writing – original draft, Writing – review & editing. **Ioannis Triantafyllou:** Software, Writing – review & editing.

Declaration of competing interest

None.

Appendix A. Survey instrument

Librarians' perceptions of student plagiarism

In your experience, classify instances of student plagiarism by intent (unintentional or intentional).

In your experience, how often do students commit the following types of plagiarism?

1 = Never to 5 = Very often (for all 11 items).

TYP1. Copying text (words, expressions, sentences, paragraphs) or ideas without (correctly) citing the author or the source.

TYP2. Copying and pasting segments of text without quotation marks, even if a citation referring has been added.

TYP3. Paraphrasing or summarizing text without (correctly) citing the author or the source.

TYP4. Submitting another student's work (e.g., assignment or paper) with or without her/his knowledge of permission.

TYP5. Submitting or presenting a group work assignment as individual academic work.

TYP6. Buying an essay or other written academic work from professional essay writing services or other third parties.

TYP7. Copying from other students or collaborating with other students without the lecturer's approval during exams or classroom assessments.

TYP8. Copying a substantial amount of content from a source, regardless of whether there is proper citation and referencing of the source.

TYP9. Using incorrect citation or bibliographic referencing, which does not provide accurate information about the original sources.

TYP10. Self-plagiarism in terms of presenting or reusing any their previous (published) work or assignment without attribution.

TYP11. Copying media (images, audio, or video) from other sources and presenting them in their own academic work or project without attribution.

In your experience, how often do students commit plagiarism for each of the following reasons?

1 = Never to 5 = Very often (for all 12 items).

R1. Foreign language problems.

R2. Time constraints, poor time management or lack of time management skills.

R3. Academic pressure and high student workload.

R4. Unawareness or poor knowledge of the concept and the forms of plagiarism.

R5. Lack of plagiarism implications or penalties and/or absence of plagiarism regulations and policy of the academic institution.

R6. Lack of adequate academic writing and research skills, including proper use of sources and proper ways of citing, referencing, quoting, paraphrasing, and summarizing and/or use of critical thinking.

R7. Difficulties in managing the research process.

R8. Pressure or desire to achieve high levels of performance and good grades.

R9. Lack of respect for academic rules and course requirements, including losing interest in learning.

R10. Teaching factors, including negative students' attitudes towards academic and teaching staff or class.

R11. Factors related to student personality, personal values and attitudes or cultural issues.

R12. Temptation and opportunities due to unlimited access to digital materials and/or to the wide use of essay writing web services.

Librarians' perceptions of the role of academic library in plagiarism prevention and detection

To what extent do you agree or disagree with the role of academic libraries in the:

1 = Strongly disagree to 5 = Strongly agree (for all 5 items).

ROLE1. Prevention of student plagiarism through awareness and provision of information.

ROLE2. Provision of student support, counselling, and guidance to avoid plagiarism.

ROLE3. Promotion of information literacy skills and ethical use of information through plagiarism education instruction.

ROLE4. Plagiarism detection in students' written academic work.

ROLE5. Participation in the development of plagiarism and academic integrity institutional policy.

Librarians' current practice in curbing students' plagiarism

As academic library professionals, how often do you participate in the following activities to inform and support students against plagiarism?

1 = Never to 5 = Very often (for all 9 items).

PREV1. Recommend information sources to students on their topic of interest.

PREV2. Answer questions about plagiarism-related issues.

PREV3. Provide information on copyright issues, academic integrity, and the ethical use of information.

PREV4. Inform about the consequences of plagiarism.

PREV5. Inform about institutional policy and procedures.

PREV6. Provide support to improve citation and referencing skills.

PREV7. Give advice on how to avoid plagiarism with the provision of anti-plagiarism techniques.

PREV8. Assist in the preparation of assignments/essays.

PREV9. Provide support to students of a suspected plagiarism incident.

In which of the following ways do you participate in informing and raising awareness among students to prevent plagiarism?

Multiple choice: check all that apply

AWRM1. Library help desk

AWRM2. Educational material on the library's website/portal

AWRM3. Phone calls

AWRM4. Information and resources on the library's website/portal

AWRM5. Online communication through digital library services (e.g., live chats)

AWRM6. Emails

AWRM7. Use of social media (marketing) tools

AWRM8. Discussion forums and newsgroups

AWRM9. Other traditional marketing techniques (flyers, posters, brochures etc.)

AWRM10. None of the above

AWRM11. Other (please specify)

How often do you participate in the following education/training activities for students?

1 = Never to 5 = Very often (for all 7 items).

EDU1. Instruction on academic ethical code and the ethical use of information.

EDU2. Information literacy instruction.

EDU3. Citation instruction (teaching of citation methods, bibliography, and in-text referencing).

EDU4. Academic writing training, including paraphrasing and summarizing.

EDU5. Teaching research methods.

EDU6. Teaching (digital) source information management, including research database searching skills, information retrieval and synthesis.

EDU7. Project and research data management tools training.

Which of the following anti-plagiarism instruction delivery methods do you use?

Multiple choice: check all that apply.

INDM1. Workshops or seminars (synchronous method of education).

INDM2. Guides and manuals.

INDM3. Lectures.

INDM4. Instructional/educational material and workbooks.

INDM5. Webinars, web tutorials or videos (asynchronous method of education).

INDM6. Semester course.

INDM7. Course Management Systems (CMSs).

INDM8. Online discussion boards or blogs.

INDM9. None of the above.

INDM10. Others (please specify).

How often do you use plagiarism detection software for:

1 = Never to 5 = Very often (for all 2 items).

DET1. Checking assignments.

DET2. Checking materials submitted to the institutional repository.

Librarians' collaborative efforts with the academic community to combat students' plagiarism

As a librarian, how often do you participate in the following

collaborative activities with the academic community to combat student plagiarism?

1 = Never to 5 = Very often (for all 6 items).

COL1. Meetings with faculty.

COL2. Collaboration with faculty for anti-plagiarism courses.

COL3. Collaboration with faculty for the development of anti-plagiarism assignments.

COL4. Provide support for the detection of plagiarism.

COL5. Provide instruction to faculty on plagiarism and the ethical use of information.

COL6. Participation in the development of the plagiarism policy.

Librarians' challenges in the prevention and detection of students' plagiarism

What are the greatest difficulties or challenges you face as academic librarians in your involvement in combating the phenomenon of plagiarism?

Multiple choice: check all that apply.

CHAL1. Collaboration issues with faculty and administration.

CHAL2. Lack of adequate knowledge and/or skills.

CHAL3. Time constraints.

CHAL4. Institutional policy for library involvement in plagiarism controlling.

CHAL5. Communication difficulties with students.

CHAL6. Other (please specify).

Note: For Sections 1–6 respondents were offered an additional free-text option to comment and indicate more information regarding their responses.

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